

# Physics of `sim_symmetric.jsx`: Rigid-Body Dynamics in a Harmonic Thermophoretic Trap

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## Abstract

This note sets out the model solved by the interactive simulator `Theory/sim_symmetric.jsx`. A rigid body of mass  $M$  and inertia tensor  $\mathbf{I}$  moves through a quiescent gas under gravity. It feels a thermophoretic force and torque that are linear in the local log-temperature gradient  $\nabla \ln T$ , plus linear translational and rotational drag. The temperature field is a phenomenological harmonic trap. Each of the four linear-response tensors is a constant symmetric  $3 \times 3$  matrix in the body frame. The 13-dimensional state (position, velocity, unit quaternion, body angular velocity) is integrated with fixed-step fourth-order Runge–Kutta. We give the equations exactly as implemented, derive the equilibrium and the small-oscillation behaviour, and list the modelling assumptions and their consequences.

## 1 Frames and state variables

We use two frames: the *lab frame*  $\{\hat{\mathbf{e}}_x, \hat{\mathbf{e}}_y, \hat{\mathbf{e}}_z\}$ , with  $\hat{\mathbf{e}}_z$  pointing up (against gravity), and the *body frame*, which is attached to the particle and whose axes are its principal axes of inertia. The rotation matrix  $\mathbf{R} \in SO(3)$  maps body-frame components to lab-frame components:

$$\mathbf{a}_{\text{lab}} = \mathbf{R} \mathbf{a}_b, \quad \mathbf{a}_b = \mathbf{R}^T \mathbf{a}_{\text{lab}}. \quad (1)$$

The state vector  $\mathbf{s} \in \mathbb{R}^{13}$  is

$$\mathbf{s} = \left( \underbrace{\mathbf{x}}_3, \underbrace{\mathbf{v}}_3, \underbrace{q}_4, \underbrace{\boldsymbol{\omega}_b}_3 \right), \quad (2)$$

where

- $\mathbf{x}$  is the centre-of-mass position (lab frame),
- $\mathbf{v} = \dot{\mathbf{x}}$  is the centre-of-mass velocity (lab frame),
- $q = (q_0, \mathbf{q})$ , with  $\mathbf{q} = (q_1, q_2, q_3)$ , is a unit quaternion ( $|q| = 1$ ) that encodes the orientation, and
- $\boldsymbol{\omega}_b$  is the angular velocity, expressed in the body frame.

**Rotation matrix.** The quaternion determines the rotation matrix through

$$\mathbf{R}(q) = (q_0^2 - |\mathbf{q}|^2)\mathbf{1} + 2\mathbf{q}\mathbf{q}^\top + 2q_0[\mathbf{q}]_\times, \quad [\mathbf{q}]_\times = \begin{pmatrix} 0 & -q_3 & q_2 \\ q_3 & 0 & -q_1 \\ -q_2 & q_1 & 0 \end{pmatrix}. \quad (3)$$

Written out using  $|q| = 1$ , this is

$$\mathbf{R} = \begin{pmatrix} 1 - 2(q_2^2 + q_3^2) & 2(q_1q_2 - q_0q_3) & 2(q_1q_3 + q_0q_2) \\ 2(q_1q_2 + q_0q_3) & 1 - 2(q_1^2 + q_3^2) & 2(q_2q_3 - q_0q_1) \\ 2(q_1q_3 - q_0q_2) & 2(q_2q_3 + q_0q_1) & 1 - 2(q_1^2 + q_2^2) \end{pmatrix}. \quad (4)$$

**Quaternion kinematics.** The orientation evolves as

$$\dot{q} = \frac{1}{2} q \otimes (0, \boldsymbol{\omega}_b) \iff \dot{q}_0 = -\frac{1}{2} \mathbf{q} \cdot \boldsymbol{\omega}_b, \quad \dot{\mathbf{q}} = \frac{1}{2} (q_0 \boldsymbol{\omega}_b + \mathbf{q} \times \boldsymbol{\omega}_b). \quad (5)$$

The body-frame angular velocity multiplies on the right in Eq. (5). This is what makes  $\mathbf{R}$  a body-to-lab map.

**Tilt.** The simulator reports the *tilt*, defined as the angle between the body symmetry axis  $\hat{\mathbf{e}}_3^b$  and the vertical:

$$\theta = \arccos(\hat{\mathbf{e}}_z \cdot \mathbf{R} \hat{\mathbf{e}}_3) = \arccos R_{33}. \quad (6)$$

## 2 Temperature field

The gas temperature enters only through the dimensionless-per-length vector field

$$\mathbf{G}(\mathbf{x}) \equiv \nabla \ln T = \frac{\nabla T}{T}. \quad (7)$$

The simulator takes this field to be affine in position. In the lab frame,

$$\boxed{\mathbf{G}(\mathbf{x}) = \mathbf{A} \mathbf{x} + \mathbf{G}_0, \quad \mathbf{A} = \text{diag}(\alpha_r, \alpha_r, \alpha_z), \quad \mathbf{G}_0 = -\frac{Mg}{\gamma_*} \hat{\mathbf{e}}_z} \quad (8)$$

Component by component,

$$\mathbf{G}(\mathbf{x}) = (\alpha_r x, \alpha_r y, \alpha_z z - Mg/\gamma_*). \quad (9)$$

The corresponding log-temperature profile is a paraboloid with a linear tilt:

$$\ln \frac{T(\mathbf{x})}{T_0} = \frac{1}{2} \alpha_r (x^2 + y^2) + \frac{1}{2} \alpha_z z^2 - \frac{Mg}{\gamma_*} z. \quad (10)$$

- **Uniform part  $\mathbf{G}_0$ .**  $\mathbf{G}_0$  points down, so the gas is hotter below and colder above, as with a hot bottom plate. Its magnitude is fixed by the particle itself: the code sets  $|\mathbf{G}_0| = Mg/\gamma_*$  so that, for an isotropic response, the thermophoretic force at the origin exactly cancels gravity (Sec. 5). The trap is therefore always centred at  $\mathbf{x} = 0$ . Changing  $M$  or  $\Gamma$  does not move the equilibrium; it only rescales the background gradient.
- **Curvature  $\mathbf{A}$ .** The constants  $\alpha_r$  and  $\alpha_z$  (units  $\text{m}^{-2}$ ) set the radial and axial trap stiffness. They are free phenomenological parameters.
- **Normalising constant  $\gamma_*$ .** The code uses  $\gamma_* = \Gamma_{22}$ , the  $(2, 2)$  entry of the thermophoretic tensor (`Gamma[3]` in the packed storage of Sec. 3). See caveat 5 in Sec. 8.

The gas is at rest everywhere, and the temperature field does not depend on time.

### 3 Linear-response tensors

Four constant  $3 \times 3$  tensors, defined in the body frame, set how the body couples to the gas. Let  $\mathbf{v}_b = \mathbf{R}^\top \mathbf{v}$  be the body-frame velocity and  $\mathbf{G}_b = \mathbf{R}^\top \mathbf{G}(\mathbf{x})$  the gradient seen by the body. Then

$$\mathbf{F}_b^{\text{drag}} = -\Xi \mathbf{v}_b \quad \text{translational drag,} \quad (11)$$

$$\mathbf{F}_b^\Gamma = -\Gamma \mathbf{G}_b \quad \text{thermophoretic force,} \quad (12)$$

$$\boldsymbol{\tau}_b^\Gamma = -\Lambda \mathbf{G}_b \quad \text{thermophoretic torque,} \quad (13)$$

$$\boldsymbol{\tau}_b^{\text{drag}} = -\mathbf{C}_r \boldsymbol{\omega}_b \quad \text{rotational drag.} \quad (14)$$

Take  $\Gamma = \gamma \mathbf{1}$  with  $\gamma > 0$  as an example. The force  $-\gamma \nabla \ln T$  then points from hot to cold, as expected for thermophoresis.

**Symmetry by construction.** The code stores each tensor as the six independent entries of a symmetric matrix,

$$\mathbf{t} \longleftrightarrow (t_{11}, t_{12}, t_{13}, t_{22}, t_{23}, t_{33}), \quad \mathbf{t} = \begin{pmatrix} t_{11} & t_{12} & t_{13} \\ t_{12} & t_{22} & t_{23} \\ t_{13} & t_{23} & t_{33} \end{pmatrix} = \mathbf{t}^\top. \quad (15)$$

This gives  $4 \times 6 = 24$  independent response parameters. The rationale given in the code is:

- $\Xi$  and  $\mathbf{C}_r$  are resistance tensors. For low-Reynolds-number flow, the Lorentz reciprocal theorem (the hydrodynamic form of Onsager reciprocity) makes them symmetric. The second law adds positive (semi)definiteness, because the dissipated power is

$$P_{\text{diss}} = \mathbf{v}_b \cdot \Xi \mathbf{v}_b + \boldsymbol{\omega}_b \cdot \mathbf{C}_r \boldsymbol{\omega}_b \geq 0. \quad (16)$$

The simulator enforces symmetry but does *not* enforce positive-definiteness. The user must pick physically sensible values.

- $\Gamma$  and  $\Lambda$  are taken to be symmetric, motivated by Onsager reciprocity. This is a modelling choice, and its consequences are discussed in Sec. 8.
- $\Lambda$  is a pseudotensor: it maps a polar vector ( $\nabla \ln T$ ) to an axial vector (torque). Its symmetric part can be nonzero only for bodies without enough improper symmetry; in particular, its trace requires a chiral body. The default  $\Lambda = 0$  is correct for a right circular cylinder.

**Lab-frame form.** In the lab frame the tensors become orientation dependent:

$$\Gamma'(q) \equiv \mathbf{R} \Gamma \mathbf{R}^\top, \quad \Xi'(q) \equiv \mathbf{R} \Xi \mathbf{R}^\top. \quad (17)$$

An isotropic tensor  $\mathbf{t} = t \mathbf{1}$  is unchanged by this rotation.

## 4 Equations of motion

### 4.1 Translation

Newton's second law for the centre of mass, with the forces evaluated in the body frame and then rotated back to the lab frame, reads

$$M \dot{\mathbf{v}} = \mathbf{R} \left[ -\Gamma \mathbf{R}^\top \mathbf{G}(\mathbf{x}) - \Xi \mathbf{R}^\top \mathbf{v} \right] - Mg \hat{\mathbf{e}}_z. \quad (18)$$

Equivalently,

$$M\ddot{\mathbf{x}} = -\Gamma'(\mathbf{Ax} + \mathbf{G}_0) - \Xi' \dot{\mathbf{x}} - Mg \hat{\mathbf{e}}_z. \quad (19)$$

## 4.2 Rotation

Euler's equations in the principal body frame, with  $\mathbf{l} = \text{diag}(I_1, I_2, I_3)$ , are

$$\mathbf{l}\dot{\boldsymbol{\omega}}_b = -\boldsymbol{\omega}_b \times (\mathbf{l}\boldsymbol{\omega}_b) - \Lambda \mathbf{G}_b - \mathbf{C}_r \boldsymbol{\omega}_b, \quad \mathbf{G}_b = \mathbf{R}^\top \mathbf{G}(\mathbf{x}). \quad (20)$$

Written out in components, the gyroscopic term gives the familiar form

$$I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_2 \omega_3 + \tau_1, \quad I_2 \dot{\omega}_2 = (I_3 - I_1) \omega_3 \omega_1 + \tau_2, \quad I_3 \dot{\omega}_3 = (I_1 - I_2) \omega_1 \omega_2 + \tau_3, \quad (21)$$

with  $\boldsymbol{\tau} = -\Lambda \mathbf{G}_b - \mathbf{C}_r \boldsymbol{\omega}_b$ . The torques are taken about the centre of mass, and the thermophoretic force is assumed to act through that point.

## 4.3 Complete system

Together with Eq. (5), the model is the closed autonomous system

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{v}, \\ M\dot{\mathbf{v}} &= -\mathbf{R}(\Gamma \mathbf{G}_b + \Xi \mathbf{R}^\top \mathbf{v}) - Mg \hat{\mathbf{e}}_z, \\ \dot{q} &= \frac{1}{2} q \otimes (0, \boldsymbol{\omega}_b), \\ \mathbf{l}\dot{\boldsymbol{\omega}}_b &= -\boldsymbol{\omega}_b \times \mathbf{l}\boldsymbol{\omega}_b - \Lambda \mathbf{G}_b - \mathbf{C}_r \boldsymbol{\omega}_b, \end{aligned} \quad \mathbf{G}_b = \mathbf{R}(q)^\top (\mathbf{Ax} + \mathbf{G}_0).$$

(22)

The translational and rotational sectors are coupled in two ways:

1. through  $\mathbf{R}(q)$  in the force, if  $\Gamma$  or  $\Xi$  is anisotropic, and
2. through  $\mathbf{G}_b(\mathbf{x}, q)$  in the torque, if  $\Lambda \neq 0$ .

# 5 Equilibrium and small oscillations

## 5.1 Isotropic response

Take  $\Gamma = \gamma \mathbf{1}$  and  $\Xi = \xi \mathbf{1}$ , so that  $\gamma_* = \gamma$ . Then  $\Gamma' = \gamma \mathbf{1}$  for every orientation. The translational equation decouples from the rotation and is *exactly* linear:

$$M\ddot{\mathbf{x}} = -\gamma \mathbf{Ax} - \xi \dot{\mathbf{x}} + \underbrace{(-\gamma \mathbf{G}_0 - Mg \hat{\mathbf{e}}_z)}_{=0}. \quad (23)$$

Gravity is cancelled identically by the background gradient. Each axis  $i \in \{x, y, z\}$  is an independent damped harmonic oscillator:

$$\ddot{x}_i + 2\zeta_i \omega_{0,i} \dot{x}_i + \omega_{0,i}^2 x_i = 0, \quad \omega_{0,i}^2 = \frac{\gamma \alpha_i}{M}, \quad \zeta_i = \frac{\xi}{2\sqrt{M\gamma \alpha_i}}, \quad (24)$$

where  $\alpha_x = \alpha_y = \alpha_r$  and  $\alpha_z = \alpha_z$ . The origin is asymptotically stable if  $\gamma \alpha_i > 0$  and  $\xi > 0$ . The translational energy

$$E_{\text{tr}} = \frac{1}{2} M |\mathbf{v}|^2 + \frac{1}{2} \gamma [\alpha_r (x^2 + y^2) + \alpha_z z^2] \quad (25)$$

is a Lyapunov function with  $\dot{E}_{\text{tr}} = -\xi |\mathbf{v}|^2$ .

**Default parameters.** With  $M = 1$ ,  $\gamma = 5$ ,  $\alpha_r = \alpha_z = 0.5$ , and  $\xi = 2$ , all three axes have

$$\omega_0 = \sqrt{2.5} \approx 1.58 \text{ s}^{-1}, \quad \zeta \approx 0.63, \quad \omega_d = \omega_0 \sqrt{1 - \zeta^2} \approx 1.22 \text{ s}^{-1}, \quad \text{decay rate } \frac{\xi}{2M} = 1 \text{ s}^{-1}. \quad (26)$$

The motion is underdamped, with period  $2\pi/\omega_d \approx 5.1 \text{ s}$ . For example, starting from rest,  $z(t) = z_0 e^{-t} [\cos \omega_d t + \omega_d^{-1} \sin \omega_d t]$  with  $z_0 = 0.2$ . The simulator output matches this to its printed precision ( $z = 0.0815, -0.0067, 0.0012$  at  $t = 1, 2, 5 \text{ s}$ ).

## 5.2 Anisotropic response

For a general  $\Gamma$ , the equilibrium at fixed orientation  $q$  satisfies

$$\Gamma'(\mathbf{A}\mathbf{x}^* + \mathbf{G}_0) = -Mg \hat{\mathbf{e}}_z \implies \mathbf{x}^* = Mg \mathbf{A}^{-1} \left[ \frac{\hat{\mathbf{e}}_z}{\gamma_*} - \Gamma'^{-1} \hat{\mathbf{e}}_z \right]. \quad (27)$$

The trap centre lies at the origin only when  $\hat{\mathbf{e}}_z$  is an eigenvector of  $\Gamma'$  with eigenvalue  $\gamma_*$ . Otherwise the equilibrium is displaced, and the displacement depends on orientation. Near equilibrium the effective stiffness matrix is

$$\mathbf{K} = \Gamma' \mathbf{A}. \quad (28)$$

$\mathbf{K}$  is symmetric only when  $\Gamma'$  and  $\mathbf{A}$  commute, which holds automatically when  $\alpha_r = \alpha_z$ . If they do not commute, the restoring force has a nonzero curl ( $\nabla \times \mathbf{F} \neq 0$ ). It is then a non-conservative “curl force” that can drive circulating orbits, so  $E_{\text{tr}}$  no longer works as a Lyapunov function.

## 6 Rotational dynamics

With  $\Lambda = 0$  there is *no orientational restoring torque*: every orientation is neutrally stable. Take isotropic rotational drag  $\mathbf{C}_r = c\mathbf{I}$  and an axisymmetric body with  $I_1 = I_2 \equiv I_\perp$ . Euler’s equations can then be solved in closed form. The spin about the symmetry axis decays exponentially,

$$\omega_3(t) = \omega_3(0) e^{-ct/I_3}. \quad (29)$$

The transverse component  $\omega_\perp \equiv \omega_1 + i\omega_2$  obeys

$$\dot{\omega}_\perp = (i\Omega(t) - c/I_\perp) \omega_\perp, \quad \Omega(t) = \frac{I_3 - I_\perp}{I_\perp} \omega_3(t), \quad (30)$$

which integrates to

$$\omega_\perp(t) = \omega_\perp(0) \exp \left[ -\frac{ct}{I_\perp} + i \frac{(I_3 - I_\perp)I_3}{I_\perp c} \omega_3(0) (1 - e^{-ct/I_3}) \right]. \quad (31)$$

So  $\omega_b$  performs a damped body-frame precession and falls to zero. The rotational energy  $\frac{1}{2} \omega_b \cdot \mathbf{l} \omega_b$  decreases at the rate  $c|\omega_b|^2$ . The orientation then *freezes* at whatever value it has reached.

**Default parameters.** With  $\mathbf{l} = \text{diag}(0.02, 0.02, 0.008)$  and  $c = 0.5$ , the spin-down times are  $I_\perp/c = 0.04 \text{ s}$  and  $I_3/c = 0.016 \text{ s}$ . Both are much shorter than the translational time scale. The total transverse rotation angle is only about  $|\omega_\perp(0)| I_\perp/c \approx 0.02 \text{ rad}$ , so the tilt stays at its initial  $30^\circ$  to the displayed precision.

**Nonzero  $\Lambda$ .** At the trap centre, a body with  $\Lambda \neq 0$  still feels the torque  $-\Lambda \mathbf{R}^\top \mathbf{G}_0 = (Mg/\gamma_*) \Lambda \mathbf{R}^\top \hat{\mathbf{e}}_z$ , because  $\mathbf{G}_0 \neq 0$ . The body therefore generically does not come to rest. A chiral body, for example, can be driven into steady rotation, with the thermophoretic torque balanced by rotational drag.

## 7 Numerical method

- **Integrator.** Classical fixed-step RK4 on  $\dot{\mathbf{s}} = \mathbf{f}(\mathbf{s})$ , with  $\Delta t = t_{\text{end}}/N$ . The default is  $t_{\text{end}} = 20 \text{ s}$  and  $N = 1200$ , which gives  $\Delta t = 1/60 \text{ s}$ .
- **Quaternion normalisation.**  $q$  is renormalised inside every right-hand-side evaluation and again after every step. This removes the drift off  $|q| = 1$  that the unconstrained RK4 update would otherwise accumulate.
- **Initial conditions.** The initial orientation is a rotation by  $\theta_0$  about  $\hat{\mathbf{e}}_x$ :  $q(0) = (\cos \frac{\theta_0}{2}, \sin \frac{\theta_0}{2}, 0, 0)$ . The user also sets  $\mathbf{x}_0$ ,  $\mathbf{v}_0$ , and  $\boldsymbol{\omega}_{b,0}$ .
- **Stability.** Explicit RK4 needs  $|\lambda|\Delta t \lesssim 2.8$  for real, negative eigenvalues  $\lambda$ . The fastest default mode is the axial spin-down, with  $c/I_3 = 62.5 \text{ s}^{-1}$ , giving  $|\lambda|\Delta t \approx 1.04$ . This is stable, though the fast rotational transient is resolved by only a few steps. Making  $I_3$  smaller or  $c$  larger by a factor of about 3 would push the integration past the stability limit.

## 8 Modelling assumptions and caveats

1. **Linear response, constant coefficients.** All four tensors are independent of position, temperature, pressure, and velocity. Real thermophoretic coefficients depend on the local gas state (for example,  $T$ ,  $p$ , and the Knudsen number) and vary across the cell.
2. **Missing cross-couplings.** The full  $6 \times 6$  resistance matrix of a rigid body has an off-diagonal translation–rotation block, which is nonzero for chiral bodies and for bodies whose reference point is not the centre of reaction. The model omits it. It likewise omits any force or torque proportional to higher gradients of  $T$ .
3. **Symmetric  $\Lambda$  excludes alignment torques.** Consider a polar but achiral body, such as a cone or a Janus particle, with symmetry group  $C_{\infty v}$  and axis  $\hat{\mathbf{n}}$ . The most general  $\Lambda$  allowed by this symmetry is purely *antisymmetric*, giving the aligning torque  $\boldsymbol{\tau} \propto \hat{\mathbf{n}} \times \nabla \ln T$ . Imposing  $\Lambda = \Lambda^\top$  forbids this torque. A symmetric  $\Lambda$  describes chirality-type couplings instead. For example,  $D_\infty$  (helix-like) bodies give  $\Lambda = \text{diag}(a, a, b)$ .
4. **Phenomenological trap.**  $\alpha_r$  and  $\alpha_z$  are free parameters. If  $T$  instead obeyed source-free steady conduction with  $\kappa \propto T^n$ , then  $\nabla \cdot (T^{n+1} \nabla \ln T) = 0$ , which implies

$$\nabla^2 \ln T = -(n+1) |\nabla \ln T|^2. \quad (32)$$

At the trap centre this requires  $2\alpha_r + \alpha_z = -(n+1)(Mg/\gamma)^2 < 0$  for  $n > -1$ . So a constant isotropic  $\gamma$  acting on a purely conductive field cannot confine the particle along all three axes at once; this is an Earnshaw-type obstruction. The harmonic trap should therefore be read as an effective description that absorbs whatever provides confinement in the experiment, such as spatially varying coefficients, boundaries, or convection.

5. **Choice of  $\gamma_*$ .** The code sets  $\gamma_* = \Gamma_{22}$  (`Gamma[3]`). But gravity is along  $\hat{e}_z$ , and for an upright body with diagonal  $\Gamma$  the component that cancels gravity is  $\Gamma_{33}$  (`Gamma[5]`). The two coincide for isotropic  $\Gamma$ , including the defaults. For  $\Gamma = \text{diag}(\Gamma_{11}, \Gamma_{22}, \Gamma_{33})$  with  $\Gamma_{22} \neq \Gamma_{33}$ , the equilibrium of an upright body moves to

$$z^* = \frac{Mg}{\alpha_z} \left( \frac{1}{\Gamma_{22}} - \frac{1}{\Gamma_{33}} \right). \quad (33)$$

6. **Positivity not enforced.** Nothing in the code prevents indefinite  $\Xi$  or  $C_r$ . Such values would inject energy and are unphysical.
7. **Principal axes.**  $\mathbf{l}$  is diagonal in the same body frame in which the response tensors are specified. If the principal axes of inertia do not coincide with the frame in which the tensors are known, the tensors must first be rotated into the principal frame.

## A Default parameters

| Quantity                          | Symbol                      | Default (as in code)                      |
|-----------------------------------|-----------------------------|-------------------------------------------|
| Mass                              | $M$                         | 1 kg                                      |
| Gravity                           | $g$                         | 9.81 m s <sup>-2</sup>                    |
| Inertia                           | $\mathbf{l}$                | diag(0.02, 0.02, 0.008) kg m <sup>2</sup> |
| Translational drag                | $\Xi$                       | 2 1                                       |
| Thermophoretic force              | $\Gamma$                    | 5 1                                       |
| Thermophoretic torque             | $\Lambda$                   | 0                                         |
| Rotational drag                   | $C_r$                       | 0.5 1                                     |
| Trap curvatures                   | $\alpha_r, \alpha_z$        | 0.5, 0.5                                  |
| Initial position                  | $\mathbf{x}_0$              | (0.5, 0.3, 0.2) m                         |
| Initial velocity                  | $\mathbf{v}_0$              | 0                                         |
| Initial body angular velocity     | $\boldsymbol{\omega}_{b,0}$ | (0, 0.5, 2) rad s <sup>-1</sup>           |
| Initial tilt (about $\hat{e}_x$ ) | $\theta_0$                  | 30°                                       |
| Duration, steps                   | $t_{\text{end}}, N$         | 20 s, 1200                                |

Note that the README lists  $\Xi = 5 \text{ 1}$ , but the code default is 2 1. With  $\xi = 5$ , the translational damping ratio would be  $\zeta \approx 1.58$ , which is overdamped, rather than 0.63.